

Box-Cox Lambda as a Redshift-Resolved Shape Statistic for Galaxy Colour Distributions: Advantages over Johnson SU Parameterisation

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Abstract

We argue that the Box-Cox power parameter λ is a superior redshift-resolved shape statistic for galaxy colour distributions compared to the two shape parameters (a , b) of the Johnson SU family. Although the Johnson SU distribution is flexible and encompasses Normal and log-Normal limits, its shape parameters are strongly correlated under maximum-likelihood optimisation, producing numerical instability when tracking smoothly evolving distributions across redshift bins. Box-Cox λ is a single interpretable parameter — the power that best normalises a positive distribution — with direct physical meaning: $\lambda = 1$ implies additive (Normal) scatter, $\lambda \rightarrow 0$ implies multiplicative (log-Normal) scatter, and $\lambda < 0$ implies a distribution sharply bounded on one side, consistent with an absorbing state such as a consolidated red sequence. We demonstrate using $u-r$ colour distributions for 165,488 GAMA DR4 galaxies in seven redshift bins ($z = 0.002-0.35$). Box-Cox λ increases monotonically from -0.20 to $+2.70$ (Spearman $r = 1.000$, $p < 0.001$) while JSU parameter a jumps non-monotonically between -11.7 and $+20.7$ across the same bins. A bootstrap null test confirms the λ trend exceeds sampling noise at every bin.

1. Introduction

Parametric fitting of galaxy property distributions is common for characterising populations and tracking redshift evolution. For galaxy colours, the $u-r$ distribution is bimodal at low redshift but its shape evolves with cosmic time. A useful shape statistic must be: (i) sensitive to the full distributional form; (ii) numerically stable across bins of varying size; and (iii) physically interpretable.

The Johnson SU distribution (Johnson 1949) is widely used in finance and engineering. Its two shape parameters a and b control skewness and kurtosis. In practice we find that maximum-likelihood estimates of a and b are strongly correlated — the likelihood surface is nearly flat along a ridge in (a, b) space — causing the optimiser to find numerically distant solutions for statistically equivalent data.

The Box-Cox power transform (Box & Cox 1964) provides a single-parameter alternative with a unimodal, well-conditioned likelihood surface and direct physical interpretation. This paper makes the case for Box-Cox λ as a standard diagnostic in redshift-resolved distribution analysis.

2. Statistical Background

2.1 Johnson SU Distribution

The JSU family is defined by $z = a + b \sinh^{-1}((x - \xi)/\text{scale})$, $z \sim N(0,1)$. Shape parameters a (skewness-related) and b (kurtosis-related, $b > 0$) are estimated by maximum likelihood. When the true distribution is close to symmetric, many (a, b) pairs fit equally well, and small perturbations in the data cause the optimiser to jump between equivalent solutions — the artefact demonstrated in Section 4.

2.2 Box-Cox Power Transform

The Box-Cox transform is defined as:

$$y(\lambda) = (x^\lambda - 1) / \lambda \quad (\lambda \neq 0)$$

$$y(0) = \log(x) \quad (\lambda = 0, \text{ by continuity})$$

The parameter λ is estimated by maximising the profile log-likelihood of $y \sim N(\mu, \sigma^2)$ over λ . The likelihood surface is smooth and unimodal for sample sizes $N \gg 100$. Physical interpretation:

$\lambda \approx 1 \rightarrow$ approximately Normal; additive scatter

$\lambda \rightarrow 0 \rightarrow$ log-Normal; multiplicative scatter

$\lambda < 0 \rightarrow$ left tail suppressed; one-way absorbing boundary

For $u-r$ colour, $\lambda < 0$ at low redshift corresponds to a consolidated red sequence acting as an absorbing state: galaxies quench but do not re-blue, producing a distribution sharply bounded on the red side.

2.3 Bootstrap Null Test for Physical Evolution

To distinguish physical evolution from sampling noise: 500 subsamples of the same size as each redshift bin are drawn with replacement from the full galaxy pool (ignoring redshift), λ is fitted to each, and the 2.5th–97.5th percentiles form the 95% null envelope. If observed λ lies outside this envelope the variation is inconsistent with the null hypothesis. Monotonic trend is assessed with Spearman r and Kendall τ , both non-parametric.

3. Data

We use GAMA DR4 (Driver et al. 2022), StellarMassesLambdarv24, with quality cuts $0.5 < u-r < 4.0$, $\log M_* > 8.0$, $0.002 < z < 0.35$: 165,488 galaxies in seven equal-width redshift bins ($\Delta z = 0.05$; $N = 3,900\text{--}34,000$ per bin). GAMA is >98% spectroscopically complete to $r < 19.8$, minimising photometric redshift scatter that would blur bin boundaries.

[Cite GAMA DR4, stellar masses paper, spectroscopic completeness.]

4. Results

4.1 JSU Instability Demonstrated

Table 1 shows all shape parameters by redshift bin. Skewness decreases smoothly from +0.90 to −0.82 across the full redshift range. Box-Cox λ increases smoothly from −0.20 to +2.70. JSU parameter a , however, jumps from −2.76 to −11.68 to +20.68 across the first three bins — a factor of >30 variation while skewness changes by only 0.75. The distributions at $z = 0.075$ and $z = 0.125$ differ modestly in skewness (+0.26 vs −0.15) and λ

(+0.43 vs +1.13), yet a changes sign and magnitude dramatically. This is numerical, not physical: the JSU likelihood ridge allows many equally valid (a, b) solutions. Figure 1 shows the contrast side by side.

Table 1. Shape parameters by redshift bin (GAMA DR4, $N = 165,488$)

z	N	JSU a	JSU b	Box-Cox λ	Skewness	Kurt
0.026	3,932	-2.763	+2.210	-0.200	+0.899	+0.824
0.075	16,579	-11.677	+5.925	+0.431	+0.263	-0.750
0.125	29,589	+20.679	+7.171	+1.125	-0.148	-1.046
0.175	32,999	+9.776	+3.566	+1.694	-0.387	-0.897
0.225	27,116	+3.431	+2.112	+1.995	-0.523	-0.623
0.275	33,685	+2.802	+1.752	+2.532	-0.745	-0.188
0.325	21,588	+2.836	+1.802	+2.701	-0.821	+0.069

JSU a : Spearman $r = +0.29$, $p = 0.53$ (not significant). Box-Cox λ and skewness: $|r| = 1.00$, $p < 0.001$. JSU a varies by a factor of >30 while skewness changes by only 0.75 — the instability is numerical, not physical.

[Insert Figure 1: Fig1_JSU_vs_BoxCox_Instability.png — 3 panels: JSU a , λ , skewness vs z]

4.2 Box-Cox λ Tracks Physical Evolution

Figure 2 shows Box-Cox λ and skewness against redshift with 95% bootstrap null envelopes. Both parameters lie outside the null envelope at every redshift bin. Spearman $r = +1.000$ ($p = 4.0 \times 10^{-4}$) for λ and -1.000 ($p = 4.0 \times 10^{-4}$) for skewness, confirming perfect monotonic physical evolution.

The λ trajectory is informative. At $z = 0.325$, $\lambda = 2.70$: the distribution is broader than Normal toward the blue, reflecting a mostly star-forming population with a growing red tail. By $z = 0.026$, $\lambda = -0.20$: the red sequence has consolidated and the distribution is sharply bounded on the red side. The crossing of $\lambda = 0$ near $z \approx 0.08$ may mark the epoch at which the red sequence first becomes the dominant population — a testable prediction for deeper surveys.

[Insert Figure 2: Fig2_Shape_vs_Redshift.png — 2-panel λ and skewness vs z with null]

5. Discussion

5.1 When to Use Box-Cox λ vs JSU

JSU remains useful when a closed-form PDF is required, e.g. generating synthetic realisations or computing tail probabilities analytically. For tracking distributional evolution with a scalar statistic across redshift bins, Box-Cox λ is preferable: single-parameter, well-conditioned likelihood, and physically interpretable. Skewness is equally stable but lacks the transform interpretation.

5.2 Generalisability

The argument applies to any redshift-binned analysis of unimodal-to-bimodal distributions: stellar mass functions, star-formation rate distributions, metallicity. One limitation: Box-Cox requires positive data. Distributions spanning zero (e.g. specific SFR including quenched galaxies at $sSFR = 0$) require a shift parameter, introducing an assumption. In such cases

skewness remains the most assumption-free scalar shape statistic.

6. Conclusions

1. Johnson SU parameters a and b show large non-monotonic variation across redshift bins due to correlation on a near-flat likelihood ridge, even when the underlying distribution evolves smoothly.
 2. Box-Cox λ is a single, well-conditioned, physically interpretable shape statistic that tracks the same evolution without numerical instability.
 3. Applied to 165,488 GAMA galaxies, λ increases monotonically from -0.20 to $+2.70$ across $z = 0.03\text{--}0.33$ (Spearman $r = 1.000$, $p = 4 \times 10^{-4}$), consistent with progressive consolidation of the red sequence.
 4. Bootstrap null testing confirms the trend exceeds sampling noise at every redshift bin.
 5. We recommend Box-Cox λ as the default shape statistic for redshift-resolved distribution analysis when a numerically stable, physically interpretable scalar is required.
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Data and Code Availability

GAMA DR4 data are publicly available at <https://www.gama-survey.org/dr4/>. Python scripts implementing the Box-Cox fitting, bootstrap null test, and figure generation are available on request (key dependencies: `scipy.stats.boxcox`, `numpy`, `matplotlib`).

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